Angel Patricio

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Education

Massachusetts Institute of Technology — **Cambridge, MA**
M.Eng., Electrical Engineering and Computer Science — *Expected 2027–2028*

Massachusetts Institute of Technology — **Cambridge, MA**
B.S., Artificial Intelligence and Decision Making — *2022–2026*
Coursework (*grad): Deep Learning*, Reinforcement Learning*, Statistical Learning Theory*, Nonlinear Optimization*, NLP, Real Analysis

Publications

J. Yuan, B. Le, K. Le, Y. Shi, A. Singh, R. Sharma, **A. Patricio**, R. Raskar. SONAR Web: A Platform-Agnostic Framework for Real-Time Decentralized Learning Across Heterogeneous Edge Clients. *ACM FLEdge-AI Workshop*, 2025. **Best Presentation Award.**

Experience

Selected Research Projects

On Monitorability in Looped Transformers — **aopatric/recurrent-grokking**
Independent Research — *Oct 2026–*
Studying the degree to which the increased serial depth in looped transformers impacts monitorability. See [my blog post] for my essay on why this is relevant to broader AI safety, and for subsequent updates on the results.
- Designed a baseline experiment for isolating serial computation in looped transformers on a task with a non-iterative shortcut, training variants of 3 architectures on n-hop modular addition base p, building on the (Nanda et al. 2023) paper on progress measures for grokking.
- Found all grokking models learned a constant-time shortcut rather than an interative algorithm, and that among grokked looped transformers there were *multiple learned strategies* for using the loop in computation. Follow-up in progress with a task that offers no shortcut.

Intact: Interpretable Activations for Reward Hacking — **aopatric/intact**
Independent Project — *Aug 2026–*
Built an experiment CLI and Python API with AI coding tools for collecting and manipulating activations from an existing reward hacking setup. Designed system for efficient sampling and writing of activations for downstream experiments in steering, probing, and patching.
- Designed a system to sample rollouts from vLLM within 4e-5 relative error to PyTorch, grade them for reward hacking with upstream's built-in grader, and store the activations from specified hook points as fp16 Parquet.
- Reproduced upstream's hack rate with the new tooling (73.7% vs. 79%). Wrote probing demonstration and accompanying documentation. Library is in beta with updates coming over time. Short writeup on the project and its epistemic status [here].

The Edge of Stability for LLM Post-Training — **aopatric/post-training-EoS**
Course Project, MIT 6.7960 Deep Learning — *Fall 2025, Revisited 2026*
Explored whether the [Edge of Stability] result extends to LLM post-training. Built new tooling after a revisit to strengthen my result.
- Identified a noisy proxy for measuring local loss curvature and an error in applying gradient clipping. Rebuilt tooling for GPU-native Hessian eigenvalue estimation in PyTorch with AI coding tools, including fixing a 3.4% relative error per step from fp32 accuracy.
- Used new tooling to observe EoS results in an MLP, and observed weak signal in models from the Pythia suite on an instruction tuning task. Reported result as inconclusive rather than null, and wrote a revisit [blog post].

Academic Research Experience

A Benchmark for Spatial Reasoning with Origami — **Cambridge, MA**
MIT Computer Science and AI Lab (CSAIL), Solar-Lezama Group — *Feb 2025–Aug 2025*
Helped design an inverse-origami benchmark for spatial reasoning; acknowledged (non-author) in OrigamiBench (ICML LM4Plan '26).
- Conducted a literature review on spatial-reasoning evaluation for both neuro-symbolic and neural-only models. Discussed findings and ideas at weekly meetings. Including presenting a demo of the benchmark before exiting the project.
- Extended an existing inverse-origami Gym environment with an .obj-to-.FOLD target pipeline (25 targets); ran search and RL baselines for comparison, and reported findings to the group.

SONAR Web: Real-Time Decentralized Learning Across Heterogeneous Edge Clients — **Cambridge, MA**
MIT Media Lab, Camera Culture Group — *Nov 2024–Feb 2025*
Built the first clients for privacy-preserving decentralized training across devices, later extended by the team and published.
- Wrote peer-to-peer communication logic for Python and JavaScript clients over the group's WebRTC layer. Streamed weights in chunks to bypass the 16MB WebRTC limit, and proposed a universal JSON weight format; validated training across up to 12 peers on 4 topologies.
- Wrote core logic in PyTorch and TensorFlow.js to train ResNets on subsets of MedMNIST with federated learning and weight averaging.

Masked Language Modeling for Genomics — **Cambridge, MA**
MIT Institute for Medical Engineering and Science (IMES), Edelman Lab — *Apr 2024–Aug 2024*
Scaled the lab's DistilBERT masked-LM pipeline for single-cell expression from 1M in Tabula Sapiens to 30M+ samples in CELLxGENE.
- Adapted data preprocessing to CELLxGENE including updated embeddings and moved training to multi-GPU cloud instances. Cut epoch time from over an hour to about 15 minutes, and trained several model checkpoints.
- Used HF Transformers and Accelerate (PyTorch DDP) on Vast.ai; built tooling to launch and log WandB sweeps in one call, as well as a Streamlit frontend for inference on user-provided data.